
AGENTDESCENT: GRADIENT DESCENT WHERE THE PARAMETERS ARE AGENTS

A PARALLEL, ASYNCHRONOUS, MERGE-BASED FRAMEWORK
FOR SELF-EVOLVING AGENT ARTIFACTS

PREPRINT

Danyang Chen

`birfychen@gmail.com`

<https://github.com/Birfy/agentdescent>

August 2026

ABSTRACT

Self-improving agent systems evolve their own skills, prompts, and harnesses, but largely through a *serial* loop that bounds improvement throughput at $O(1/T_{\text{iter}})$. **AgentDescent** transplants the distributed-training stack onto this problem: artifacts as parameters, diffs with attached evidence as gradients, a version-vectored ledger as the parameter server, and merge decisions as optimizer steps. Because diffs do not add, aggregation becomes a discrete-space optimization — conflict resolution, fusion, and statistical acceptance under a per-artifact Beta posterior. Eighteen published algorithms port onto the engine as plug-ins and run unchanged under serial, synchronous, and barrier-free scheduling.

Measured on live models, synchronous parallelism gives a median $1.36\times$ end-to-end speedup across eleven algorithms (11 of 11, $p=0.001$), while removing the round barrier buys a further $1.31\times$ where the gate is expensive and nothing across eleven compact ports. Merging supplies a third source of speedup that selection cannot: a round’s k concurrent proposals are validated once rather than k times, and validation is where a live-model loop spends its critical path. We derive merging’s precondition in closed form — the engine proposes only from failed rollouts, so a near-saturated artifact starves the merge step however finely its keys are cut — and on the workload that condition prescribes, merging spends 11–32% fewer model calls than a serial lineage and best-of- N selection at a pinned rollout budget, with no quality difference detectable at the resolution six seeds afford. Code, workloads, and per-result reproduction commands are open source.

Keywords Self-evolving agents · Parallel and asynchronous execution · Diff merging · LLM agents · Distributed optimization

1 Introduction

A growing body of work shows that large-language-model agents can improve their own artifacts: prompts [4, 7], in-context playbooks [41], skill libraries [32, 43], agentic workflows [11, 40], and their own code [25, 36, 39]. Most of these systems share an execution structure inherited from the Gödel machine [27]: a serial improvement loop in which one candidate modification is proposed, evaluated, and accepted or rejected before the next is considered. If one iteration takes T_{iter} wall-clock seconds, improvement throughput is bounded at one accepted change per T_{iter} , independent of available compute. Notable exceptions parallelize at the *population* level — FunSearch and AlphaEvolve run asynchronous samplers over island archives [24, 26] — but their candidates remain independent: concurrent work is selected among, not merged.

Distributed deep learning encountered the same bottleneck and resolved it with a now-standard stack: data-parallel workers computing gradients concurrently, parameter servers aggregating them [19], staleness-tolerant asynchronous

updates [23], per-parameter adaptive steps [17], and gradient surgery for conflicting updates [37]. This paper develops the thesis that the same stack, suitably reinterpreted, applies to agent self-evolution — and identifies precisely where the reinterpretation must depart from the original.

AgentDescent (Fig. 1) instantiates the mapping. A library of evolvable artifacts plays the role of the parameter tensor; a diff annotated with an *evidence card* (the rollout that motivated it) plays the role of a gradient; a git-backed, version-vectored ledger plays the role of the parameter server; and the optimizer step is an *aggregator* that merges concurrent diffs. N workers roll out tasks against snapshots of the library and propose diffs in parallel; an asynchronous merger aggregates them, targeting $O(N/T_{\text{iter}})$ improvement throughput.

The mapping is productive precisely because of where it breaks. *Gradients add; diffs do not*. Two gradients can always be summed; two textual edits to the same artifact may be complementary, redundant, or contradictory. Aggregation is therefore not averaging but a decision procedure — staleness filtering, conflict resolution, fusion, and a statistical acceptance test (Section 4). A second disanalogy is equally consequential: training code is not modifiable by the gradients it computes, whereas a self-evolving system can in principle rewrite its own verifier. AgentDescent therefore enforces a layered governance model in which safety-critical components are frozen (Section 6).

Contributions.

1. **A system design and a discrete-space aggregator** mapping the distributed-training stack onto parallel self-evolution — per-diff bounded staleness, conflict projection, fusion, Beta-posterior acceptance, dual-branch promotion — with a formal account of where the analogy holds and where it must break (Sections 3 and 4).
2. **An efficiency analysis identifying three sources of speedup** (Section 5): rollout overlap, barrier removal, and — specific to merge-based evolution — reduced validation cost, in which merging collapses k acceptance tests into one and *reflective merge* extends the reduction to artifacts whose concurrent proposals conflict. The three do not compose unconditionally, and we characterize the regimes in which each pays.
3. **An extensibility mechanism** under which published algorithms are ported as plug-ins rather than forks (Sections 3.3 and 7): a diff-encoding strategy plus an optional `aggregator_factory`, or a declarative policy bundle. Eighteen ports run under all three schedulers without modification.
4. **A precondition for merging, in closed form** (Eq. (5)): because the engine proposes only from failed rollouts, the fraction of rounds with two diffs to fuse is $1 - (1 - f)^N - Nf(1 - f)^{N-1}$ in the incumbent’s failure rate f and the worker count N . It retrodicts two of our null results, prescribes the workload that avoids them, and bounds fusion width from below where the trust region and the round count bound it from above.
5. **A live-model evaluation** (Section 8) across eighteen ports and four questions: an eleven-algorithm scheduler study in which synchronous parallelism wins for all eleven methods (median 1.36× end-to-end) while barrier removal pays only where the gate is expensive (6.28× there, 0.99× across compact ports) — a regime distinction we characterize rather than average over; direct observation of the validation-cost reduction as falling model-seconds per rollout at a pinned budget; quality preserved across ten of eleven ports under the most aggressive schedule; and an equal-budget merge-versus-selection comparison on a workload built to satisfy the precondition above, in which merging is the cheapest arm in both model calls and wall-clock at indistinguishable held-out quality.

2 Related Work

Self-evolving agents span prompt evolution [4, 7], context and playbook evolution [41], skill libraries [32, 43], workflow and harness search [11, 40], self-modifying code [25, 36, 39], evolutionary program search [18, 22, 24, 26, 28], and label-free self-play [2, 12, 42], with reflection [21, 29] as a shared inner proposal mechanism. Prompt optimization more broadly includes meta-prompting and search methods — APE [44], OPRO [34], DSPy [16], EvoPrompt [10] — and TextGrad [38], which also frames LLM feedback as a gradient analogue: TextGrad backpropagates textual critiques through a single computation graph, whereas AgentDescent’s “gradients” are concurrent keyed diffs merged into one lineage under statistical acceptance. Population-based training [14] and the island architectures of FunSearch and AlphaEvolve parallelize by maintaining independent candidates and selecting among them; AgentDescent instead merges concurrent edits at the diff level. Most of the systems above publish a serial outer loop; AgentDescent treats that loop as one scheduling regime among three. On the systems side, AgentDescent borrows its vocabulary from distributed training: parameter servers [19], lock-free and bounded-staleness asynchrony [1, 8, 23], per-parameter adaptivity [17], gradient surgery [37], weight averaging [33], and tensor/pipeline parallelism [13, 30]. Concurrent work on parallel self-evolution (barrier-free evolution loops; co-evolutionary verification [3]) indicates that the throughput premise is an open hypothesis; our contribution is the specific synthesis — concurrent, staleness-bounded, conflict-resolved diff-level merge over a versioned ledger — together with the measurement harness required to evaluate it.

Table 1: The correspondence between distributed training and AgentDescent. The final row is a deliberate disanalogy, enforced rather than elided.

Model training	AgentDescent
parameter tensor θ	library of Evolvable artifacts
gradient g	Diff + EvidenceCard
parameter server	git-backed, version-vectored Ledger
optimizer step	Aggregator merge decision
per-parameter adaptive step (Adam)	per-artifact Beta-posterior acceptance
bounded staleness (async SGD/RL)	per-diff lag η + rebase re-verification
EMA / weight averaging	dev/stable dual branch
training code (not self-modifiable)	L0 frozen governance layer

3 Problem Setup and System Design

3.1 Problem formulation

Let $\mathcal{A}$ denote the space of *artifact libraries*: finite keyed collections $A = \{a_k\}$ of versioned textual or executable artifacts (skills, prompts, playbooks, harness modules, file trees). An agent parameterized by A induces a policy π_A ; given a task distribution $\mathcal{D}$ and a bounded reward $r \in [0, 1]$, the objective is

$$A^* = \arg \max_{A \in \mathcal{A}} \mathbb{E}_{x \sim \mathcal{D}} [r(\pi_A(x), x)], \quad (1)$$

optimized from rollout evidence only: $\mathcal{A}$ is discrete, and no gradient of r with respect to A exists. The serial baseline — propose one candidate A' , evaluate on a held-out split D_{val} , accept if better — performs one accepted modification per iteration time T_{iter} . AgentDescent’s premise is that with N concurrent proposers and merge-based aggregation, accepted-modification throughput can approach $O(N/T_{\text{iter}})$ without sacrificing the acceptance discipline.

A *diff* d is a keyed partial edit over A , produced from a rollout by a *strategy* S that maps a natural-language proposal to operations on keys (rule slots, playbook bullets, file paths). The key structure is what makes two diffs comparable: edits on disjoint keys *fuse*; edits on the same key *conflict*. Every diff carries an *evidence card* $e = (x, \tau, r, v_{\text{base}})$ recording the task, rollout, reward, and the library version the proposer observed.

3.2 System components

Table 1 summarizes the correspondence and Fig. 1 the resulting architecture. The **ledger** is the parameter server: a git-backed store with per-artifact version vectors, compare-and-swap (CAS) commits, and a dual dev/stable branch pair; stable is promoted after K regression-free rounds on dev, an EMA-style confirmation in which a new commit restarts the counter. **Workers** hold snapshots of the library, roll out tasks, and emit (diff, evidence) pairs into a thread-safe buffer. The **aggregator** consumes the buffer and performs merge decisions (Section 4). Version vectors make the staleness of any diff computable at merge time:

$$\eta(d) = \max_{a \in \text{touched}(d)} (v_{\text{head}}(a) - v_{\text{base}}(a)). \quad (2)$$

3.3 Pluggable seams

Each component in Fig. 1 sits behind an explicit interface, selected by one keyword argument of the single entry point `evolve()`: the *agent layer* (any prompt $\rightarrow$ text callable is a completion — Claude, OpenAI-compatible endpoints, a tool-using CLI agent, or plain `run/reward/propose` functions), the *strategy* (diff encoding), the *aggregator* (aggregator_factory), the *staleness policy*, the *runtime* (synchronous or barrier-free), the *parallelism scheme* (data/tensor/pipeline [13, 30]), and the *executor and sandbox*. Selection and acceptance rules are likewise named policy classes. Section 7 uses these seams to port eighteen published algorithms without modifying the engine. The single deliberate exception is governance (Section 6): the L0 layer exposes no seam, since a self-modifying system must not be able to replace its own evaluator.

4 The Aggregator as a Discrete-Space Optimizer

Evidence cards are bucketed per artifact; when a bucket reaches batch size B (or a timeout, so cold artifacts are not starved), the aggregator executes one optimizer step, summarized in Algorithm 1: **(1) staleness filtering** by Eq. (2)

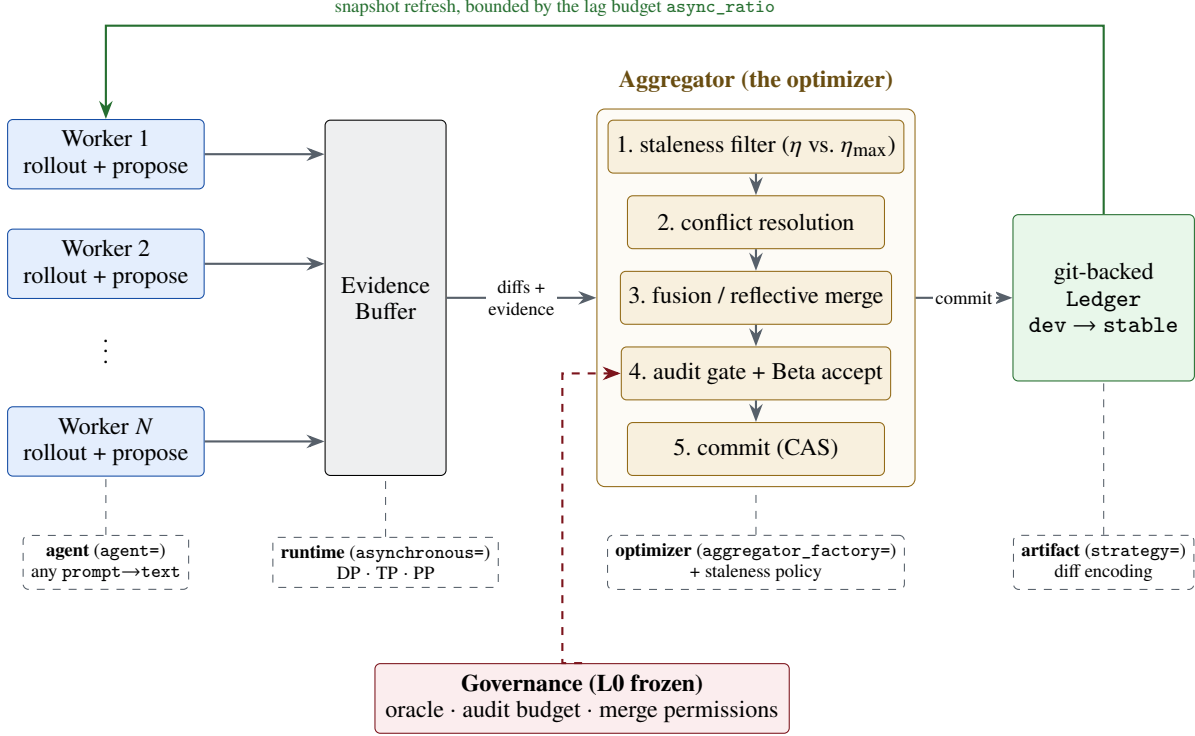


Figure 1: AgentDescent architecture. **Solid, top:** the fixed data path — N workers roll out tasks against ledger snapshots and emit diffs with evidence cards into a thread-safe buffer; a dedicated aggregator thread runs the five-stage merge pipeline and commits winners to the git-backed ledger; the green return edge is the only feedback path, bounded by the lag budget. **Dashed, bottom:** the pluggable seams of Section 3.3, each selected by one keyword argument of `evolve()`. **Red:** the frozen L0 governance layer, deliberately not a seam.

under the active policy (Section 5.2); **(2) conflict resolution** — syntactic (overlapping hunks) and semantic (contradictory operations) conflicts are detected, and contradictions are projected out in the manner of PCGrad [37], retaining the better member of each pair on a shared held-out subset; **(3) fusion** — complementary survivors are merged into one union candidate (cf. model soups [33]), with *reflective merge* handling contradictions that a keyed union cannot (Section 5.3); **(4) auditing** — merges with high blast radius or low trust are routed through an oracle with veto power; **(5) statistical acceptance and commit.** Let (s_c, f_c) and (s_b, f_b) be the candidate’s and the incumbent’s successes and failures on the held-out split D_{val} . The candidate commits (by CAS on dev) iff

$$\Pr[\theta_c > \theta_b] > 1 - \delta_v, \quad \theta_c \sim \text{Beta}(1 + s_c, 1 + f_c), \quad \theta_b \sim \text{Beta}(1 + s_b, 1 + f_b), \quad (3)$$

with independent uniform priors and the probability estimated by Monte Carlo, matching the released implementation — a per-artifact posterior test rather than a point threshold, the counterpart of per-parameter adaptivity in Adam [17]. The confidence requirement δ_v is annealed over the artifact’s version v (the learning-rate-decay analogue) and a trust region caps diff size.

One risk of this discipline deserves explicit statement: the gate reuses a single fixed D_{val} across many acceptance decisions (hundreds of proposals per run in Section 8.3), so accepted diffs are selected extreme statistics on D_{val} and their measured gains are optimistically biased — the standard adaptive-data-analysis concern. Two mitigations are in place and one is not: all quality results in Section 8 are reported on a disjoint *test* split the gate never sees; the trust region and annealed δ_v limit how much any single decision can exploit D_{val} ; but no reusable-holdout or D_{val} -rotation mechanism is implemented. We therefore measure the resulting bias directly (Section 8.3): across 33 runs the gain the gate observed on D_{val} exceeds the gain realized on test by 0.038 on average, a small but statistically detectable optimism that a reader should subtract from any D_{val} -reported figure.

A natural objection to merge-based evolution is that two individually beneficial edits may be jointly harmful. Two properties bound the risk: the acceptance test Eq. (3) scores the *fused* candidate on the full held-out split, so a harmful fusion cannot commit; and the union is a superset of the diffs that survive conflict resolution — which drops the weaker member of each contradicting pair first, so it is not a superset of every proposal. An optional *fusion tournament* additionally ranks the union against its constituents; it is disabled by default (it costs one additional held-out sweep per

candidate against a recoverable gain) but serves as the measurement instrument for the objection, reporting win rate, the losing tail, and a contested counter that records whether a fused candidate was in fact constructed and ranked — so a run in which fusion never occurred cannot be reported as a merge win. Section 8.4 reports the win rate in situ for one workload; we attach no headline to it, because it is a property of that artifact’s key granularity and proposal overlap rather than of the mechanism.

5 Efficiency Analysis: Three Sources of Speedup

Wall-clock time in a self-evolution loop decomposes into rollout time, synchronization time, and held-out validation time. AgentDescent addresses each with a distinct mechanism; the third is specific to merge-based evolution.

5.1 Overlapping I/O-bound rollouts

Agent rollouts are dominated by time spent waiting on a model endpoint, so thread-level concurrency suffices to overlap them (Section 8.1 measures $5.8\times$ at eight threads on real API calls). The synchronous data-parallel runtime shards tasks across N workers over a common snapshot and merges at a round barrier. The barrier, however, imposes a ceiling independent of N : with i.i.d. rollout latencies t_i ,

$$T_{\text{round}}^{\text{sync}} = \mathbb{E}[\max_{i \leq N} t_i] + T_{\text{gate}}, \quad (4)$$

and reasoning-model latencies are heavy-tailed — a short answer and a long deliberation are the same call — which maximizes the gap between $\mathbb{E}[\max_i t_i]$ and $\mathbb{E}[t]$. The second term is a serial region in which all workers idle while the gate evaluates.

5.2 Barrier-free execution under bounded staleness

The asynchronous runtime (Algorithm 1) removes the round structure: workers produce continuously, a dedicated merger consumes continuously, and per-rollout cost approaches $\mathbb{E}[t]$ — the no-barrier discipline of asynchronous RL systems [1, 8] transferred to self-evolution. Its cost is staleness, which Eq. (2) makes first-class and per-diff, bounded by two mechanisms. A *lag budget* ρ (`async_ratio`) limits drift at the source: a worker refreshes its snapshot once v_{head} exceeds its snapshot version by more than ρ , and backpressure forces a global resynchronization if evidence accumulates while nothing commits. A *staleness policy* disposes of stale diffs at the merger: FULL admits them unconditionally (sound only where diffs compose); GUARDED rebases within $\eta \leq \eta_{\text{max}}$ and discards beyond, in the spirit of bounded-staleness RL [8], paying for safety in discarded rollouts; REFLECTIVE always rebases and *re-verifies* — a stale diff is treated as unproven rather than invalid, and is discarded only if its claimed improvement fails to hold on the new head. Because ρ is denominated in artifact versions, its appropriate value scales with rollout cost; the runtime emits a warning when a run discards the majority of its evidence.

5.3 Reducing validation cost

The third source of speedup is unavailable to the first two mechanisms: the gate itself. Every acceptance decision costs one sweep over D_{val} , and on live models this dominates the merger’s critical path: profiled on a live GEPA/HotpotQA run ($N=4$, GLM-5.2), the merge thread’s gate time *equals* its total busy time (560.1 s of each within a 740 s process, i.e. merger occupancy 75.7%, all of it evaluation). A serial or selection-based loop with m proposals pays $m \cdot |D_{\text{val}}|$ gate evaluations. Merging changes the count: a round’s k surviving diffs fuse into one candidate, so per-round gate cost falls from $k \cdot |D_{\text{val}}|$ to $|D_{\text{val}}|$; a diff discarded by the staleness filter never reaches the gate and costs zero evaluations; and in the asynchronous runtime, cards arriving during a merge enlarge the next batch rather than triggering additional merges. The effect is visible directly in the RQ1 experiment (Fig. 5): model-seconds per rollout fall across the serial, synchronous, and asynchronous arms at a pinned rollout budget. Two costs partially offset the saving and are stated for balance: the REFLECTIVE staleness policy’s re-verification spends additional gate evaluations, and a model-written reflective merge spends one further model call per contested fusion. Residual gate cost parallelizes independently of the worker pool (`eval_concurrency`), saturating at $|D_{\text{val}}|$.

The reduction presupposes that proposals *can* fuse, which fails exactly when the artifact’s key space is too coarse. An artifact held in a single key — e.g. a prompt as one instruction slot, GEPA’s setting [4] — makes every pair of concurrent proposals a write–write conflict; a keyed union then degenerates to retaining the better single proposal, i.e. best-of- n selection at best-of- n validation cost. *Reflective merge* restores the reduction: a model is prompted to write the *union* of the contradicting proposals — one candidate preserving each proposal’s intent — and the union passes through the same acceptance test as any candidate, so the model proposes and the gate decides. Key granularity is necessary but

Algorithm 1 Barrier-free evolution (`async_evolve`); the synchronous runtime is the special case with a round barrier between the two loops. S is the strategy of Section 3.1. Backpressure has two parts: `buffer.put` blocks while the unmerged intake exceeds the lag budget, and a stall detector (evidence arriving while nothing commits) forces a global resynchronization. In the current implementation a single merger thread performs all commits, so the CAS never contends; multi-merger operation, where it would, is future work.

```

1: Worker  $i$  (in parallel,  $i = 1..N$ ):
2: loop
3:   if  $v_{\text{head}} - v_i > \rho$  then refresh snapshot  $A_i \leftarrow$  ledger head;  $v_i \leftarrow v_{\text{head}}$ 
4:   end if
5:   sample task  $x$ ; rollout  $\tau \leftarrow \pi_{A_i}(x)$ ; reward  $r \leftarrow r(\tau, x)$ 
6:    $(d, e) \leftarrow S.\text{propose}(A_i, x, \tau, r)$ ; buffer.put( $d, e$ ) ▷ blocks on backpressure
7: end loop
8: Merger (single thread):
9: loop
10:   $\mathcal{B} \leftarrow \text{buffer.drain}()$ ; bucket  $\mathcal{B}$  by artifact
11:  for each bucket with  $|\mathcal{B}_a| \geq B$  or timed out do
12:     $\mathcal{B}_a \leftarrow \{d : \text{policy}(\eta(d), \eta_{\max}) \neq \text{DISCARD}\}$ , rebasing where directed ▷ Eq. (2)
13:    resolve conflicts (keep better of each contradicting pair);  $u \leftarrow \text{fuse}(\text{survivors})$ 
14:    if survivors contradict on a key and reflective merge enabled then  $u \leftarrow$  model-written union
15:    end if
16:    if audit passes and Eq. (3) holds on  $D_{\text{val}}$  then CAS-commit  $u$  to dev
17:    end if
18:  end for
19:  promote dev  $\rightarrow$  stable after  $K$  regression-free rounds
20: end loop

```

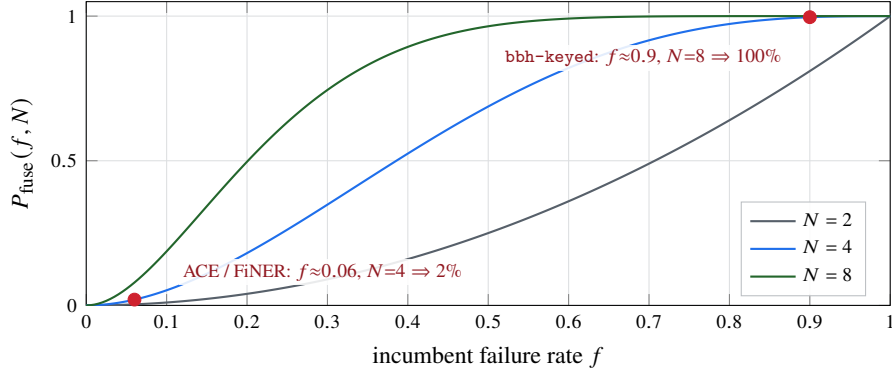


Figure 2: The fusion precondition of Eq. (5): the fraction of rounds in which at least two diffs exist, and therefore anything to fuse at all. Because the engine proposes only from *failed* rollouts, a near-saturated artifact starves the merge step regardless of how finely its keys are cut — the curve collapses at small f for every width. Red markers are the two workloads of Section 8.4: the one whose merge step never fired sits at 2%, and the one built to satisfy the condition sits at 100%. Widening N moves the curve left, which is how the equation bounds fusion width *from below*; the round count and the trust region bound it from above.

not sufficient, and the missing condition is quantifiable. The engine proposes only from a rollout that *failed* — a solved task yields no diff — so if the incumbent fails a fraction f of tasks and N workers run per round, the probability that a round produces at least two diffs, and therefore anything to fuse, is

$$P_{\text{fuse}}(f, N) = 1 - (1 - f)^N - Nf(1 - f)^{N-1}. \quad (5)$$

This is the design condition merging must satisfy before it can be compared with selection at all, and it is easy to violate: a near-saturated artifact starves the merge step no matter how finely its keys are cut. Section 8.4 measures both sides of it. It is the spatial counterpart of the REFLECTIVE staleness policy (carry intent onto a changed base; let measurement decide), and the contested counter reports whether fusion genuinely occurred, so the saving is never claimed where only selection took place.

6 Governance by Blast Radius

Artifacts are assigned to layers by a declared `blast_radius`: **L2** (skills, prompts) merges under the full asynchronous pipeline; **L1** (harnesses, verifiers) serializes in-flight changes and routes every merge through the oracle; **L0** (the oracle, audit budget, merge permissions, safety constraints) is frozen and read-only to the loop. The frozen layer enforces the second disanalogy of Table 1: without it, a self-referential loop can converge to a verifier that accepts its own output. The same layering protects executed agent code, which runs behind a frozen test suite the candidate cannot rewrite; pristine tests are overlaid after workspace materialization, so weakening them at run time is likewise ineffective.

7 Case Study: Porting Eighteen Self-Evolution Algorithms

Because every component sits behind a seam (Section 3.3), porting a published algorithm requires a small set of plugins rather than a fork. Two routes cover all eighteen ports.

Route 1: a strategy, plus an optimizer where needed. A *benchmark-faithful* port supplies a strategy encoding the paper’s artifact as keyed diffs and, only when its search differs from the default pipeline, an `aggregator_factory`. The seven faithful ports illustrate the scale of the required code: ACE [41] is a playbook strategy whose curator *is* the default aggregator; GEPA [4] substitutes a per-instance Pareto optimizer; ADAS [11] and DGM [39] plug in keep-all archives with their own parent selection; EvoSkill a bounded top- K frontier; SkillOpt a strict edit gate;¹ OpenEvolve [28] MAP-Elites islands [22]. Parent-selection and acceptance rules are extracted as named policy classes, so subsequent ports reuse them.

Route 2: a declarative policy bundle. The eleven microports and analogues (PromptBreeder, AFlow, Self-Refine, Reflexion, SICA, Gödel Agent, Voyager, SkillWeaver, Absolute Zero, R-Zero, Agent0) define no new classes: each is a `MethodPolicy` — selection rule, memory, acceptance shape, genome — over one shared runner, inheriting a common budget contract, dataset layer, and command line. A shared `--serial` flag reproduces the upstream algorithm’s own semantics (one worker, serial gate) as the control that any parallelization claim requires, and a shared rollout budget pins compared arms to equal work.

Fidelity is declared per port — *benchmark-faithful*, *mechanism microport*, *self-edit analogue*, *environment analogue*, or *inference analogue* — follows the released code where it diverges from the paper, and documents infrastructure boundaries explicitly; analogues are not to be cited as benchmark reproductions. All eighteen ports run under serial, synchronous, and barrier-free scheduling without modification, which is what permits the controlled scheduler comparison of Section 8.2.

8 Evaluation

Setup. All results are obtained from live models on real datasets: `deepseek-v4-flash` and `GLM-5.2` (reasoning models) via an OpenAI-compatible endpoint, on each algorithm’s own benchmark. Every number is produced by a named script in the repository with its exact reproduction command; absolute times depend on host and endpoint, so ratios are the reproducible quantity. We address four questions: **RQ1** — how much effective concurrency do the three mechanisms of Section 5 deliver against a faithful serial control? **RQ2** — does the speedup generalize across algorithms? **RQ3** — is learning behavior preserved under the most aggressive schedule, and how optimistic is the gate that decides it? **RQ4** — does merging accelerate without a quality penalty relative to best-of- N selection at equal budget?

Statistical conventions. Spreads across seeds are sample standard deviations. Sign tests are exact and two-sided with ties dropped, and the unit of analysis is stated with each one. No correction for multiple comparisons is applied anywhere; the paper makes on the order of a hundred pairwise comparisons, so a single nominal p near 0.05 should be read as descriptive. Where a design cannot reach significance — which is the case for every RQ4 quality comparison — we report paired confidence intervals and the minimum detectable effect instead of a p -value.

Table 2 states every configuration in full, including the two runs that produced null results and are reported as such. Three conventions hold throughout and are worth stating once, because each of them costs us something. The rollout budget is *pinned on every arm* of every comparison, so a wide arm cannot buy its speedup with extra work — the failure mode that makes an unbudgeted $N=8$ arm look eight times better. The serial control runs its gate at

¹EvoSkill and SkillOpt are ports of released systems whose upstream papers and per-port departures are identified in the repository’s port-fidelity documentation; we do not cite them here to avoid attributing a reconstruction to the wrong reference.

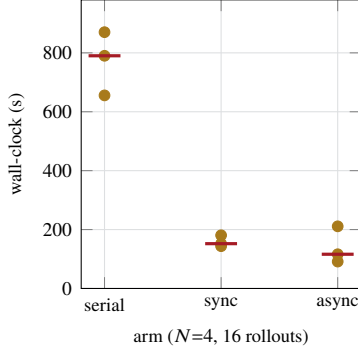


Figure 3: RQ1, the quantity the framework exists to reduce: wall-clock to spend a pinned rollout budget. GEPA on HotpotQA, GLM-5.2, 16 rollouts on every arm, three seeds, medians marked. $790 \rightarrow 152 \rightarrow 116$ s: parallelism takes the first $5.2\times$, barrier removal a further $1.31\times$, $6.8\times$ end to end.

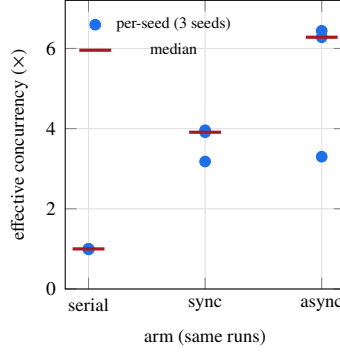


Figure 4: Where that time goes: effective concurrency (model-seconds $\div$ wall-clock). The serial control reproduces the published loop (one worker, serial gate) and lands at $0.99\text{--}1.00\times$ on every seed — a stable control property, not a single-run coincidence. Synchronous parallelism spans $3.18\text{--}3.96\times$; barrier removal spans $3.30\text{--}6.44\times$, the widest arm, and reaches it while overrunning the pinned budget (19 rollouts against 16).

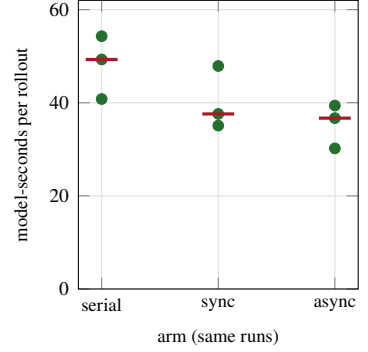


Figure 5: Validation-cost reduction in the same runs (Section 5.3): model-seconds spent per rollout at a pinned rollout budget, per seed, with medians marked. The parallel arms evaluate less per unit of work — median $49.3 \rightarrow 37.6 \rightarrow 36.7$. We do *not* attribute this to fusion: RQ1’s artifact is GEPA’s single-key instruction slot, on which a keyed union is never contested, so the collapse of Section 5.3 cannot be what fired here. The candidates are the admission cap (one merged candidate per round, so the Pareto pool grows by one instead of N) and staleness discards, which reach the gate at zero cost. Separating them needs an arm without the cap, which we have not run.

`eval_concurrency=1`, reproducing the published loop rather than a partly-parallel version of it; this makes the control slower and every speedup beside it larger, which is why the value is recorded rather than assumed. And the held-out *test* split is disjoint from the gate’s D_{val} in every experiment, so no quality figure below was ever seen by an acceptance decision.

8.1 RQ1: Effective concurrency against a faithful serial control

We first verify the premise of Section 5.1: eight threads over one pool reduce a real GLM-5.2 API workload from 48.6 s to 8.3 s ($5.8\times$ in this run; $5.8\text{--}6.5\times$ across three repeats, sub-linear under heavy-tailed latency), while the same threads on pure-Python CPU work yield $1.1\times$ — rollout overlap is real, and the GIL does not limit I/O-bound rollouts.

The main experiment ports GEPA to its own benchmark (HotpotQA [4, 35]) with 16 rollouts pinned on every arm, three seeds, and a control that reproduces the upstream loop exactly: one worker *and* a serial gate. That control lands at 0.99, 1.00, and 1.00 $\times$ overlap on the three seeds (Fig. 4), which is worth stating on its own: the property that makes it a faithful control is stable, not a fortunate single run. Against it, synchronous parallelism spans $3.18\text{--}3.96\times$ effective concurrency (median 3.91) and barrier removal spans $3.30\text{--}6.44\times$ (median 6.28); median wall-clock falls $790 \rightarrow 152 \rightarrow 116$ s (Fig. 3) — a $6.8\times$ reduction in the time to spend the same rollout budget, which is the framework’s core claim in its most direct form. Figure 5 shows the validation-cost mechanism of Section 5.3 in the same runs, and it replicates across seeds: model-seconds per rollout fall from a median 49.3 (serial) to 37.6 (synchronous) to 36.7 (asynchronous) at the pinned budget — the parallel arms not only finish sooner, they evaluate less per unit of work. Independent endpoint probes sustain $7\times$ (matched width and request length) to $10\times$ (wider, shorter requests) concurrency, so no arm is provider-limited.

Three qualifications bound these numbers. First, *the asynchronous arm is not paying the same bill*: lacking a round boundary it ran 19 rollouts against the pinned 16 on every seed, so part of its concurrency is work the other arms did not perform. Second, *it is also the least stable arm* — a $3.30\text{--}6.44\times$ spread against synchronous parallelism’s $3.18\text{--}3.96\times$. Third, and most important, *none of this is a quality claim*: median held-out test score is 0.55 (serial), 0.60 (synchronous), 0.50 (asynchronous) across the three seeds, on a 20-item split whose resolution is 0.05, so the arms are indistinguishable in quality and the fastest arm is not the best one. One synchronous cell (seed 0) also ended early on an endpoint error at 12 of 16 rollouts and is included as measured.

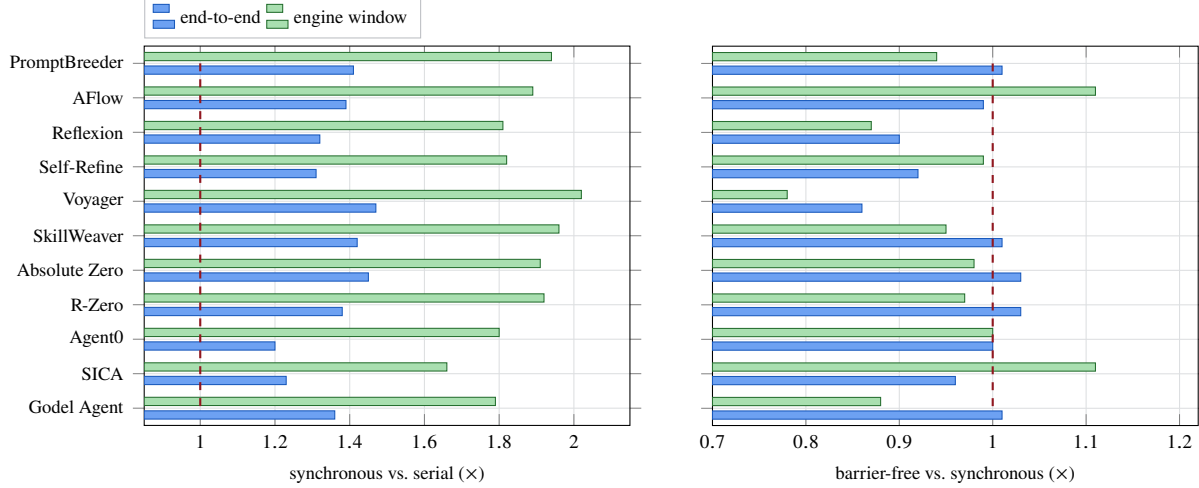


Figure 6: RQ2, per method: the two halves of the scheduler result, on the same eleven ports and the same runs. **Left:** synchronous parallelism against the serial control clears 1.0 (dashed) for all eleven methods on both clocks — unanimous, $p = 0.001$ by a two-sided sign test on each. **Right:** removing the barrier, against synchronous, straddles 1.0: 5 methods faster and 5 slower end-to-end ($p = 1.00$), 2 faster and 8 slower in the engine window ($p = 0.11$). The two panels use different x ranges and neither starts at zero, so bar *lengths* are not comparable across panels — read the positions against the dashed line. This is the regime distinction of Section 5.3 made visible — these compact ports have little heavy-tailed rollout latency or gate cost for asynchrony to hide, unlike the gate-heavy workload of Fig. 4.

These figures supersede a single-run measurement of $1.85 \times / 3.22 \times$ on a different model reported in an earlier draft. Two differences account for the gap and neither is a correction of the other: this run uses GLM-5.2 rather than deepseek-v4-flash, and its parallel arms run the gate at `eval_concurrency=16`. The second is the more instructive: with the gate itself parallelized, the round barrier costs much less, which both raises the synchronous arm and shrinks what barrier removal has left to recover — exactly the regime dependence that Section 8.2 finds across eleven algorithms. An earlier draft additionally reported a width-8 variant with a 28% model-call saving; it is withdrawn because its serial arm ran with a concurrent gate (overlap $1.31 \times$), violating the control definition above.

8.2 RQ2: Generality across algorithms

To test whether the RQ1 result generalizes beyond GEPA, the same scheduler comparison — serial, synchronous, barrier-free — is applied to all eleven declarative ports on GLM-5.2 at an equal candidate budget and width 2, varying only the scheduler (three paired seeds per method; 99 observations). The results cut both ways, and we report both. *Synchronous parallelism generalizes:* paired over $n=33$ runs it outperforms the serial arm by a median $1.36 \times$ end-to-end and $1.89 \times$ within the engine window. The win is unanimous across methods — 11 of 11 on both the end-to-end and engine-window medians, a two-sided sign test giving $p = 0.001$ in each case. *Barrier removal does not:* the asynchronous arm aggregates to $0.99 \times$ end-to-end and $0.97 \times$ engine-window relative to synchronous, and the per-method record is indistinguishable from chance — 5 methods faster, 5 slower, 1 tied end-to-end ($p = 1.00$), and 2 faster against 8 slower in the engine window ($p = 0.11$, i.e. if anything a slight cost). The reconciliation with RQ1’s $1.31 \times$ wall-clock gain from barrier removal is the regime: barrier removal pays in proportion to how much heavy-tailed rollout latency and gate cost there is to hide, which GEPA’s long rollouts and expensive Pareto gate maximize and these compact ports mostly lack; asynchrony helps time-to-quality only when a useful commit can land before a synchronous barrier would. Two caveats: this matrix was measured on the implementation preceding the ports’ declarative restructuring (mechanisms and budgets unchanged by design; a rerun on current code is pending), and its per-run JSON is retained as a summarized report rather than raw. We report it despite the caveats because it is the only multi-algorithm evidence in either direction, and it bounds the asynchronous claim honestly. Figure 6 shows the per-method record behind both halves: every bar in the left panel clears 1.0, and the right panel straddles it.

8.3 RQ3: Quality preservation under the aggressive schedule

Throughput is meaningful only if learning is preserved, so quality is measured under the most aggressive configuration (barrier-free, 8 workers, FULL staleness). Figure 7 reports the eleven declarative ports at three seeds each. Read precisely: ten of eleven improve their three-seed mean test score beyond their domain’s noise floor; Reflexion’s $+0.016$ on

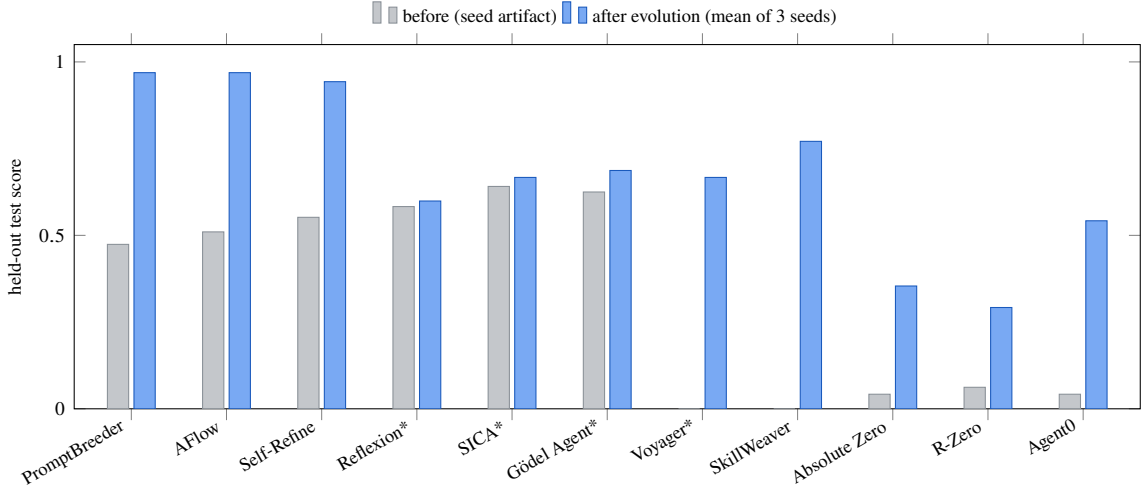


Figure 7: Held-out test score before and after evolution for the eleven microports and analogues, each run against deepseek-v4-flash under the *most aggressive* configuration — barrier-free async pipeline, 8 workers, Full staleness, 80 roll-outs per seed, 3 seeds (465–2,069 model calls per seed). “After” bars are three-seed means; asterisked methods improved on two of three seeds (the rest on all three), and Reflexion’s gain is within its domain’s ± 0.02 noise floor. Between 3 and 10 of the 240 proposals per method (80×3 seeds) commit — the gate admits only strict held-out improvements, so sparse acceptance is the shape of the mechanism, not a low yield. Bars are comparable only within a domain; per-seed spreads are tabulated in the repository documentation.

GSM-Hard [5, 9] sits within that domain’s ± 0.02 floor and should be read as no change; and four methods (Reflexion, SICA, Gödel Agent, Voyager) improve on two of three seeds rather than all three. Gains are bounded by each benchmark’s headroom — the three GSM8K ports converge to 0.94–0.97 and differ in cost rather than quality. What this establishes is that the most aggressive schedule the framework offers — no barrier, eight workers, staleness admitted unconditionally — does not stop published algorithms from learning on their own benchmarks; a paired serial-schedule arm at equal budget would additionally price asynchrony *relative to serial*, and is left to future work. Among the benchmark-faithful ports, three representative single-seed results establish end-to-end operation (not effect sizes): ACE improves FiNER-139 [20] validation $0.719 \rightarrow 0.766$ over 120 rollouts while curating a ten-bullet playbook; DGM’s best archived child repairs held-out vendored bugs at 0.906 versus its seed agent’s 0.844 under a frozen pytest oracle (a compact SWE-bench-style setting [15]); SkillOpt improves a hard SearchQA [6] subset $0.053 \rightarrow 0.211$ with 3 of 43 proposed edits accepted. One port produced no quality number at all: ADAS’s benchmarks were either saturated (MGSM) or cost-prohibitive per call (GPQA), and we state this rather than average over it. The highest-level interface (`evolve_skill`) on 40 HotpotQA rows raises held-out exact match from 0.167 to 0.583 over 338 model calls, committing one proposal.

Is the gate that decides this optimistic? It reuses one fixed D_{val} across every decision, so an accepted diff is a selected extreme statistic on that split (Section 4). The 33 runs above measure the resulting bias directly, since each records both its D_{val} trajectory and its disjoint test score: mean val gain $+0.395$ against mean test gain $+0.357$, a mean gap of -0.038 (median -0.031 , s.d. 0.075), with test gain short of val gain in 20 of 33 runs and above it in 7 — a two-sided sign test on the 27 non-tied runs gives $p = 0.019$. That test treats the 33 runs as independent, which they are not: they are 11 ports $\times$ 3 seeds, and seeds within a port share a benchmark, a D_{val} , and a reward granularity. Taking the port as the unit instead gives 8 negative port means against 2 positive and 1 tied, $p = 0.109$ — not separable. The run-level result is also fragile: dropping any one of five of the eleven ports moves p above 0.05, and dropping SkillWeaver alone — whose test split has eight items, so one task is 0.125 — moves the mean gap from -0.038 to -0.026 . What we claim is therefore the direction and the magnitude, not significance: the optimism is consistently negative, about a tenth of the mean gain, concentrated in the coarsest-reward ports, and too small to change the sign of any result here, where every quality figure is reported on test.

8.4 RQ4: Merge-of- N versus best-of- N selection at equal budget

Throughput cannot distinguish merging from sampling-and-selecting, since N independent forks occupy N workers equally well; the discriminating quantity is held-out quality at an equal rollout budget. The largest such run to date — HotpotQA on GLM-5.2, three arms $\times$ three seeds at nine rollouts per seed, budget pinned in rollouts, reflective merge

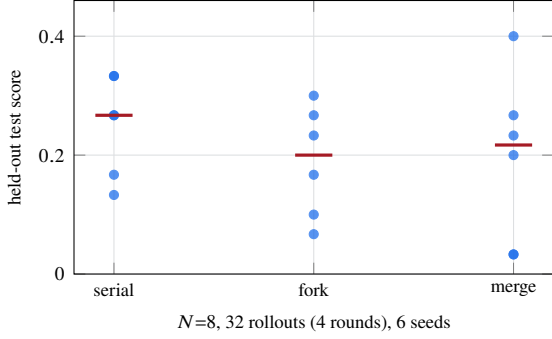


Figure 8: Fusion width 8. Per-seed test scores with medians marked. The three arms overlap; the merge arm has the *widest* spread (s.d. 0.142 against 0.084 and 0.093).

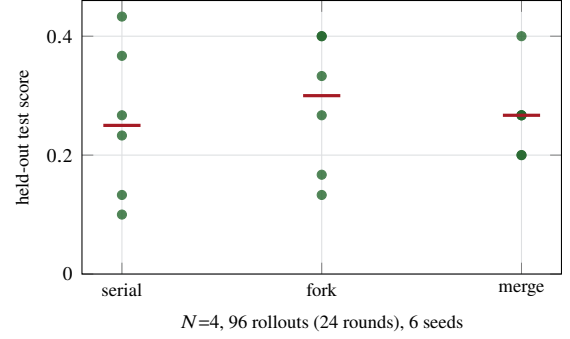


Figure 9: Fusion width 4, matched to the trust region, at a $3\times$ budget, six seeds. The arms still overlap, but the merge arm is the *tightest* (s.d. 0.073 against 0.129 and 0.115) — the ordering that width 8 reverses.

enabled so that contradicting proposals survive to fusion; 1,298 model calls (29 failing at the endpoint) and 11.3 hours of model time — does not separate the arms: merge-of-3 attains the highest median test score together with the widest spread, the intervals overlap, and the framework’s own separation test declines to declare a winner. The diagnosis is the useful part, and it is what produced Eq. (5): at this budget the fused union was constructed twice in the entire experiment — most merges saw a single surviving diff — so the experiment does not measure a merging deficit; it measures a regime in which merging rarely fired. It also confirms the key-granularity condition of Section 5.3 from the opposite direction: without reflective merge, the merge arm on this single-key artifact would have reduced to best-of- N selection. We therefore tested that diagnosis on a *multi-key* artifact: ACE’s playbook on FiNER-139, keyed per bullet, so two workers editing different bullets compose by construction. Fusion still did not fire — over a 24-rollout, four-worker run the aggregator held **2 tournaments and contested none**, one with a single candidate and none with contradicting candidates. The cause was not key granularity but Eq. (5): that artifact already scored 0.938 on the gate’s split, so $f \approx 0.06$, and at $N=4$ the predicted fraction of rounds with anything to fuse is 2%. Two uncontested tournaments in six rounds is what a 2% rate looks like.

Equation (5) also prescribes the fix, so we built the workload it calls for. BBH [31] supplies tasks a strong model still fails, and KeyedRules supplies an artifact keyed by the category a failure diagnosis falls into (output format, method, verification, common errors). Crossing them needed no engine change — only a new WorkLoad, because dataset and strategy are independent seams (Section 3.3). On GLM-5.2 the seed artifact scores 0.100 on test, so $f \approx 0.9$, and at $N=8$ Eq. (5) predicts that essentially every round should have something to fuse.

It does. Over a 32-rollout run the aggregator held **3 contested tournaments** — the first in this project in which fusion actually ran — while the artifact improved from 0.100 to 0.400 on test, so the workload supplies both the failure rate and the headroom the question needs. The fused candidate **won 1 of the 3**, at a mean gain of -0.115 with two losses (worst -0.250); two of the three fell below the artifact they started from. On this workload, fusing a round’s diffs was usually *worse* than keeping the better single one — and the acceptance gate refused exactly those, which is the safeguard of Section 4 behaving as designed rather than an argument against it.

With fusion firing on demand, the comparison the question has been waiting for becomes possible. We ran it at three seeds, and then — because the three-seed reading was favorable to merging — at three more. The second half did not replicate the first, and the combined result is the one we report: six seeds $\times$ three arms on bbh-keyed, 32 rollouts pinned on every arm, $N=8$, GLM-5.2 (16,124 model calls, 28.3 hours of model time).

Merging is the cheapest arm at equal quality. The framework’s premise is throughput — accepted improvement per unit wall-clock — so the claim merging has to meet is non-inferiority in quality at lower cost. The six seeds deliver the cost half cleanly and the quality half only weakly. Quality: the paired merge–fork difference is $+0.005$ with a 95% interval of $[-0.186, +0.196]$ at $N=8$ and -0.017 with $[-0.133, +0.100]$ at $N=4$. Those intervals are wider than the arm means themselves (0.19–0.28), so what the design can detect is a penalty of roughly 0.16–0.26 test-score points and nothing smaller. We report the intervals rather than a p -value because at six seeds with ties the sign test cannot reach $\alpha = 0.05$ at all — its smallest attainable p is 0.0625 at $N=8$ and 0.125 at $N=4$ — so a non-significant result there carries no information. Cost, by contrast, is measured cleanly: merge is the cheapest arm in model calls — 11% fewer than serial and 22–32% fewer than fork. This is the validation-cost mechanism of Section 5.3 showing up as money (Fig. 10): the merge arm reaches the same held-out score having run the gate fewer times.

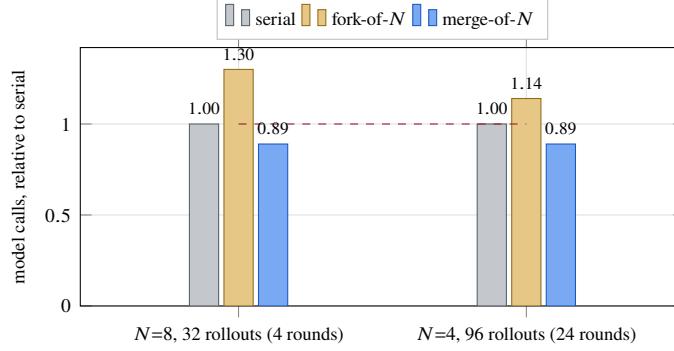


Figure 10: What the arms of Table 3 cost, at equal held-out quality and an equal rollout budget, normalized to the serial control in each configuration (absolute counts: 772/1004/685 and 2070/2363/1832). Merging is the cheapest arm in both configurations and by the same margin — 11% below serial and 22–32% below fork-of- N — because a round’s surviving diffs are validated once rather than once each. Selection has no such saving available: N independent forks must each be scored, which is why the fork arm sits *above* the serial control despite doing the same number of rollouts.

We report no wall-clock advantage over fork, and an earlier draft’s claim of one was wrong. The fork arm’s runs were executed sequentially (`--fork-concurrency 1`), so the wall-clock we compared against was the *sum* of N forks rather than what N concurrent forks would take. On the same runs the longest single fork is 320–376 s against merge’s 1,127–1,176 s: given N workers, fork-of- N finishes a pinned rollout budget *faster* than merge-of- N , because each fork is a short serial lineage with no round barrier and no shared gate. That is the honest comparison and it cuts against us on this axis. Merging’s cost advantage is in model calls, where the gate is amortized; wall-clock at equal worker count belongs to selection.

Two bounds travel with the quality column. Seeds 0–2 alone put merge’s median at 0.267 against fork’s 0.100 and seeds 3–5 reversed it; we report the pooled six rather than the favorable half. The decision to run the second three was made *after* seeing the first three and because they were favourable, which is data-dependent stopping: the six-seed figures are therefore descriptive, and we do not attach inferential weight to them beyond the intervals above. And six seeds on a 30-task split detect only large effects, so what is established is the absence of a large quality penalty.

Fusion width is a design parameter, and Eq. (5) does not fix it alone. The precondition is satisfied almost equally by $N=4$ (99.6% of rounds have something to fuse) and $N=8$ (100%), but the two differ in what they do to the rest of the loop: at a fixed rollout budget $N=8$ halves the number of rounds, and therefore the opportunities to accumulate, while making each fused union an eight-way jump that the trust region of Section 4 exists to prevent. Width should therefore be set at the smallest N that clears Eq. (5), not the largest available — a concrete design rule the equation yields once the round count is priced in. Re-running at $N=4$ and a $3\times$ budget (24 rounds), also at six seeds, moves the mechanism in the predicted direction: the mean per-seed fusion gain rises from -0.135 to -0.045 , and the merge arm’s seed-to-seed standard deviation falls from 0.142 — the *worst* of the three arms — to 0.073, the *lowest*, against serial’s 0.129 and fork’s 0.115. None of these variance ratios is separable at six seeds (every F is far inside the two-sided critical value at 5, 5 degrees of freedom), and the ordering reverses between the two widths, so we report it as an ordering with a mechanism attached and not as an effect. Quality remains indistinguishable within intervals as wide as those above, and merge remains the cheapest arm (1832 calls against 2070 and 2363).

That variance ordering is the observation we would most like to see tested properly, and six seeds cannot test it. It has a mechanism to its name — merging averages several workers’ evidence into one artifact, where selection keeps whichever single lineage the gate happened to prefer — but the evidence has three problems we state rather than argue past: no variance ratio is significant at $n=6$, the ordering *reverses* between the two widths, and the ratio halved when we went from three seeds to six. A quantity that flips across configurations and shrinks as the sample grows is what noise looks like. We record it as a hypothesis with a mechanism, not as a finding.

The fusion counters, across both configurations. At $N=8$ the aggregator held 10 contested tournaments over six seeds and the fused candidate won 2 (20%); at $N=4$ it held 21 and won 4 (19%), at a mean per-seed gain of -0.045 against width 8’s -0.135 . Fusion loses more contests than it wins in both — and this is why the design works: the acceptance test refuses exactly those losses, so carrying a usually-worse union costs one gate sweep and nothing else, while the wins are kept. Non-inferior quality at a 20% fusion win rate is the gate doing its job, not a result in spite of it. The negative control holds throughout: across eighteen control runs the serial and fork arms held 823 tournaments and contested *none*, as their construction requires.

The summary is therefore narrower than an earlier draft of this paper claimed: **on this workload merging costs 11–32% fewer model calls than the alternatives at a pinned rollout budget, and we cannot detect a quality difference at the resolution six seeds afford.** The cost result is the measured one; the quality result is an interval too wide to exclude penalties smaller than about 0.16. Merging is not shown to improve quality, and it does not win on wall-clock.

9 Limitations

AgentDescent is a research reference implementation, and four boundaries delimit what its evaluation establishes. (i) *Effect sizes.* The quality results establish direction and rule out large penalties rather than measuring magnitudes: six seeds on a 30-task split in RQ4, three seeds for the RQ1 ladder and the eleven-port study, one seed for each benchmark-faithful port. Merging is shown non-inferior and cheaper, not superior, and the variance ordering of Section 8.4 is reported as a direction rather than a ratio. (ii) *Scale.* The system is a single Python process — one merger thread (measured at 75.7% occupancy at $N=4$, all of it gate evaluation), thread workers, a local git ledger. Multi-machine operation, a contended CAS path, and a worker sweep beyond $N=8$ are outside this paper; the merger-saturation point they would locate is the natural next system question, and Section 5.3’s profile says where to look for it. (iii) *Holdout reuse.* The gate reuses one fixed D_{val} across all acceptance decisions (Section 4); test splits are disjoint and the resulting optimism is measured at 0.038 (Section 8.3), but no reusable-holdout or D_{val} -rotation mechanism is implemented. (iv) *Reruns and transfer.* The RQ2 matrix was measured on the implementation preceding the ports’ declarative restructuring and awaits a rerun; difficulty parameters calibrated on one model did not transfer under a model swap, whereas benchmark-intrinsic difficulty did. Every number reported here is generated by a named script, and measurements that proved irreproducible or favorably defined — including, in an earlier draft, a width-8 experiment withdrawn for a flawed control — were re-measured and corrected in place. We regard that discipline as part of the contribution: the evaluation of a self-improving system should satisfy the standard its own acceptance gate imposes on diffs.

10 Conclusion

The distributed-training stack admits a coherent counterpart in agent self-evolution once its necessary disanalogy — diffs do not add — is taken seriously and merging is constructed as a discrete-space optimizer. Efficiency decomposes into three sources with different evidential status: rollout overlap and synchronous parallelism generalize across algorithms (median 1.36× end-to-end over eleven ports, all eleven improving); barrier removal is a regime-dependent win, reaching a median 6.28× effective concurrency on a gate-heavy workload while showing no aggregate gain across eleven compact ones; and reduced validation cost — merging collapses k acceptance tests into one — is visible directly as falling model-seconds per rollout at a pinned budget. Eighteen published algorithms adapt as plug-ins rather than forks, and run under all three schedulers unmodified — which is what makes the scheduler a controlled variable instead of a property of each paper’s own loop. Merging itself we place on a testable footing: its precondition is stated in closed form — enough failures per round that two diffs coexist — so a workload can be built to satisfy it on demand, and Eq. (5) yields the design rule that fusion width should be the smallest N that clears it. On the workload that condition prescribes, merging is an *acceleration at no measurable quality cost*: indistinguishable from both a serial lineage and best-of- N selection at six seeds, for 11–32% fewer model calls. That its fused candidates lose more contests than they win is the design working rather than failing — the acceptance gate makes carrying a usually-worse union cost one held-out sweep and nothing more, which is precisely why k proposals can be validated once instead of k times. The result we would most like to see tested next is the one this arrangement suggests: that the product of merging is predictability, and that a self-evolution loop should be judged on accepted improvement per unit of validation rather than per proposal.

Acknowledgments

OpenEvolve and the eleven microports and analogues were contributed by cyanneko.

Reproducibility

All code is available at <https://github.com/Birfy/agentdescent> (MIT license; `pip install agentdescent`; the core engine has no required dependencies). Generating scripts, per result: thread overlap — `examples/efficiency.py --only gil`; RQ1 and the live merger profile of Section 5.3 — `bench/matrix_run.py --rows gepa`; RQ2 — `bench/candidate_methods.py`, measured on the pre-restructuring implementation (summarized report retained, raw per-run JSON not; rerun pending); RQ3 — each port’s own command line over

examples/_method_runner.py, the entry point that accepts the `--staleness` full configuration used, with the resolved configuration printed as one JSON line per run; the `evolve_skill` case — examples/skill_evolution.py; RQ4 — bench/baselines_run.py `--dataset bbh-keyed --fusion-tournament`, whose workload (BBH crossed with KeyedRules), width and budget sweeps, and fusion counters were added for this paper and ship with it. Every port supports `--dry-run` (no API key or model call) and carries an offline test suite; datasets are retrieved from canonical sources by a dependency-free data layer.

References

- [1] ROLL Flash: Accelerating RLVR and agentic training with asynchrony. arXiv preprint, 2025. Preprint; identifier omitted pending verification.
- [2] Agent0: Unleashing self-evolving agents from zero data via tool-integrated reasoning. arXiv preprint, 2025. Preprint; identifier omitted pending verification.
- [3] Self-evolving agent skills via co-evolutionary verification (CoEvoSkills). arXiv:2604.01687, 2026. Concurrent work; official code unreleased at the time of writing.
- [4] Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, et al. GEPA: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*, 2025.
- [5] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [6] Matthew Dunn, Levent Sagun, Mike Higgins, V Ugur Guney, Volkan Cirik, and Kyunghyun Cho. SearchQA: A new Q&A dataset augmented with context from a search engine. *arXiv preprint arXiv:1704.05179*, 2017.
- [7] Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Prompt-breeder: Self-referential self-improvement via prompt evolution. *arXiv preprint arXiv:2309.16797*, 2023.
- [8] Wei Fu, Jiaxuan Gao, Xujie Shen, Chen Zhu, Zhiyu Mei, Chuyi He, Shusheng Xu, Guo Wei, Jun Mei, Jiashu Wang, et al. AReaL: A large-scale asynchronous reinforcement learning system for language reasoning. *arXiv preprint arXiv:2505.24298*, 2025.
- [9] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *International Conference on Machine Learning*, pages 10764–10799, 2023.
- [10] Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations*, 2024.
- [11] Shengran Hu, Cong Lu, and Jeff Clune. Automated design of agentic systems. *arXiv preprint arXiv:2408.08435*, 2024.
- [12] Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. R-Zero: Self-evolving reasoning LLM from zero data. *arXiv preprint arXiv:2508.05004*, 2025.
- [13] Yanping Huang, Youlong Cheng, Ankur Bapna, Orhan Firat, Dehao Chen, Mia Chen, HyoukJoong Lee, Jiquan Ngiam, Quoc V Le, Yonghui Wu, et al. GPipe: Efficient training of giant neural networks using pipeline parallelism. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [14] Max Jaderberg, Valentin Dalibard, Simon Osindero, Wojciech M Czarnecki, Jeff Donahue, Ali Razavi, Oriol Vinyals, Tim Green, Iain Dunning, Karen Simonyan, Chrisantha Fernando, and Koray Kavukcuoglu. Population based training of neural networks. *arXiv preprint arXiv:1711.09846*, 2017.
- [15] Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? *arXiv preprint arXiv:2310.06770*, 2023.
- [16] Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into self-improving pipelines. In *International Conference on Learning Representations*, 2024.
- [17] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

- [18] Joel Lehman, Jonathan Gordon, Shawn Jain, Kamal Ndousse, Cathy Yeh, and Kenneth O Stanley. Evolution through large models. In *Handbook of Evolutionary Machine Learning*, pages 331–366. Springer, 2023.
- [19] Mu Li, David G Andersen, Jun Woo Park, Alexander J Smola, Amr Ahmed, Vanja Josifovski, James Long, Eugene J Shekita, and Bor-Yiing Su. Scaling distributed machine learning with the parameter server. In *11th USENIX Symposium on Operating Systems Design and Implementation (OSDI 14)*, pages 583–598, 2014.
- [20] Lefteris Loukas, Manos Fergadiotis, Ilias Chalkidis, Eirini Spyropoulou, Prodromos Malakasiotis, Ion Androutsopoulos, and Georgios Paliouras. FiNER: Financial numeric entity recognition for XBRL tagging. In *Proceedings of ACL*, 2022.
- [21] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [22] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.
- [23] Feng Niu, Benjamin Recht, Christopher Ré, and Stephen J Wright. HOGWILD!: A lock-free approach to parallelizing stochastic gradient descent. In *Advances in Neural Information Processing Systems*, volume 24, 2011.
- [24] Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025.
- [25] Maxime Robeyns, Martin Szummer, and Laurence Aitchison. A self-improving coding agent. *arXiv preprint arXiv:2504.15228*, 2025.
- [26] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M Pawan Kumar, Emilien Dupont, Francisco J R Ruiz, Jordan S Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024.
- [27] Jürgen Schmidhuber. Gödel machines: Self-referential universal problem solvers making provably optimal self-improvements. *arXiv preprint cs/0309048*, 2003.
- [28] Asankhaya Sharma. OpenEvolve: an open-source evolutionary coding agent. <https://github.com/codelion/openevolve>, 2025.
- [29] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [30] Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. Megatron-LM: Training multi-billion parameter language models using model parallelism. *arXiv preprint arXiv:1909.08053*, 2019.
- [31] Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V Le, Ed H Chi, Denny Zhou, and Jason Wei. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.
- [32] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.
- [33] Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, pages 23965–23998, 2022.
- [34] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *International Conference on Learning Representations*, 2024.
- [35] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W Cohen, Ruslan Salakhutdinov, and Christopher D Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of EMNLP*, 2018.
- [36] Xunjian Yin, Xinyi Wang, Liangming Pan, Xiaojun Wan, and William Yang Wang. Gödel agent: A self-referential agent framework for recursive self-improvement. *arXiv preprint arXiv:2410.04444*, 2024.
- [37] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. *Advances in Neural Information Processing Systems*, 33:5824–5836, 2020.

-
- [38] Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Zhi Huang, Carlos Guestrin, and James Zou. TextGrad: Automatic “differentiation” via text. *arXiv preprint arXiv:2406.07496*, 2024.
 - [39] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin gödel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025.
 - [40] Jiayi Zhang, Jinyu Xiang, Zhaoyang Yu, Fengwei Teng, Xionghui Chen, Jiaqi Chen, Mingchen Zhuge, Xin Cheng, Sirui Hong, Jinlin Wang, et al. AFlow: Automating agentic workflow generation. *arXiv preprint arXiv:2410.10762*, 2024.
 - [41] Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, et al. Agentic context engineering: Evolving contexts for self-improving language models. *arXiv preprint arXiv:2510.04618*, 2025.
 - [42] Andrew Zhao, Yiran Wu, Yang Yue, Tong Wu, Quentin Xu, Matthieu Lin, Shenzhi Wang, Qingyun Wu, Zilong Zheng, and Gao Huang. Absolute zero: Reinforced self-play reasoning with zero data. *arXiv preprint arXiv:2505.03335*, 2025.
 - [43] Boyuan Zheng, Michael Y Fatemi, Xiaolong Jin, Zora Zhiruo Wang, Apurva Gandhi, Yueqi Song, Yu Gu, Jayanth Srinivasa, Gaowen Liu, Graham Neubig, et al. SkillWeaver: Web agents can self-improve by discovering and honing skills. *arXiv preprint arXiv:2504.07079*, 2025.
 - [44] Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. Large language models are human-level prompt engineers. In *International Conference on Learning Representations*, 2023.

Table 2: Every configuration behind Section 8. Budget is pinned in rollouts on all arms of a comparison unless noted. Flags not listed take each script’s defaults, which every run prints in its header and records in its result JSON; Section 10 names the script per experiment. Model names are the endpoint’s identifiers.

Experiment	Model	Workload	Arms, width	Budget, seeds	Gate, staleness, notes
Thread overlap (Section 8.1)	GLM-5.2	8 concurrent API calls of matched request length	8 threads vs. 1	3 repeats	Pure-Python CPU work run through the same harness as the GIL control
RQ1 (Section 8.1)	GLM-5.2	HotpotQA via the GEPA port; --fetch 80, --val-cap 8	serial / sync / async, $N=4$	16 rollouts per arm; seeds 0–2	Gate at eval_concurrency=16 on the parallel arms, =1 on the serial control; --async-ratio 3; --rounds 9999 so the budget binds, not the port’s own iteration default; --reflective-merge on every arm, so it is not a difference between them; reasoning tokens disabled
RQ2 (Section 8.2)	GLM-5.2	each port’s own benchmark, 11 ports	serial / sync / async, $N=2$	equal candidate budget; 3 paired seeds per cell (99 runs)	GUARDED staleness throughout — this entry point exposes no staleness flag; measured on the implementation preceding the ports’ declarative restructuring
RQ3 (Section 8.3)	deepseek- v4-flash	each port’s own benchmark, 11 ports	barrier-free only, $N=8$	80 rollouts per seed; seeds 0–2	FULL staleness, the most permissive policy — stale diffs admitted unconditionally; 465–2,069 model calls per seed
RQ4, null 1 (Section 8.4)	GLM-5.2	HotpotQA; artifact is a single instruction slot	serial / fork / merge, $N=3$	9 rollouts per seed; seeds 0–2	Reflective merge on. 1,298 calls, 11.3 h model time. Reported as a null: the fused union was built twice in the whole experiment
RQ4, null 2 (Section 8.4)	GLM-5.2	FiNER-139 via the ACE playbook, keyed per bullet; --pool 2000, --val-cap 32	serial / fork / merge, $N=4$	24 rollouts; seed 0	Reflective merge and fusion tournament on. Reported as a null, and Eq. (5) explains it: the seed artifact already scored 0.938 on the gate’s split, so $f \approx 0.06$
RQ4, width 8 (Section 8.4)	GLM-5.2	bbh-keyed: BBH dyck_languages crossed with KeyedRules; 60 train / 30 val / 30 test	serial / fork / merge, $N=8$	32 rollouts per arm (4 rounds); seeds 0–5	Keyed union (DefaultFusion) with the fusion tournament and self-verification on; <i>not</i> reflective merge — ReflectiveFusion records every trial unranked, so the contested counts reported below could not exist under it. 16,124 calls, 28.3 h model time
RQ4, width 4 (Section 8.4)	GLM-5.2	as above, same splits	serial / fork / merge, $N=4$	96 rollouts per arm (24 rounds); seeds 0–5	As above. The width matched to the trust region, at a 3× budget so the round count rises rather than falls

Table 3: Merge-of- N against best-of- N selection and a serial control at an equal rollout budget, on a workload built to satisfy Eq. (5), in two configurations. *Test* is a split no gate in any arm ever saw; *s.d.* is across seeds. *Oracle* is the best fork *on test*, an upper bound that requires the answer to select, against which the fork row reports what fork-and-select can actually ship. *Calls* is the per-seed median. Wall-clock is deliberately not tabulated: the fork arm ran its N sub-runs sequentially, so its recorded wall-clock is a sum rather than what N concurrent forks would take, and the arms of a seed shared one endpoint. Model calls are unaffected by both and are the cost currency we report.

	arm	test min/med/max	mean	s.d.	oracle	calls
$N=8, 32$	serial	0.133 / 0.267 / 0.333	0.250	0.084	—	772
	fork-of-8	0.067 / 0.200 / 0.300	0.189	0.093	0.267	1004
	merge-of-8	0.033 / 0.217 / 0.400	0.194	0.142	—	685
$N=4, 96$	serial	0.100 / 0.250 / 0.433	0.256	0.129	—	2070
	fork-of-4	0.133 / 0.300 / 0.400	0.283	0.115	0.333	2363
	merge-of-4	0.200 / 0.267 / 0.400	0.267	0.073	—	1832

Both blocks: six seeds. Upper: 4 rounds per run. Lower: 24 rounds per run.